

Midterm Report

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Part A: Machine Learning Component

Data Preparation & Environment Setup

Requirements:

- Prepare the environment and data for modeling using the [Ames Housing](#) library for the dataset and appropriate packages for preprocessing, modeling, and visualization

Implementation and Decisions

- R was the programming language
- Key packages used:
 - [Rsampl](#) for data splitting
 - [Recipes](#) for feature engineering
 - [Dplyr](#) and [tidyr](#) for data manipulation
 - [Ggplot2](#) for visualization
 - [H2O](#) for modeling
 - [Xgboost](#) for further machine learning tasks

Tools & Technologies:

- R with a version of 4.3.1 or higher
- H2O which is a cluster-based ML framework
- Packages mentioned above

Data Pre-Processing

Requirements

- Load and split dataset
- Prepare features for modeling

Implementation & Decisions

- Loaded the dataset and utilized stratified sampling to maintain the distribution of the target feature [Sale_Price](#)
- Feature engineering was completed to reduce sparsity
- Recipes were prepared for H2O compatibility

Tools & Technology

- [AmesHousing](#) dataset
- [Recipes](#) package

- H2O data frames

Base Learner Models

Requirements

- Choose the appropriate base learning models (regularized regression, random forest, gradient boosting, XGBoost)
- Implement and choose the hyperparameter values

Regularized Regression (Base Model)

- **Decision:** Utilized as the base model to be compared to the other base learners and super learner
- Built with the `h2o.glm` function

Random Forest

- **Decision:** Utilized a good mix of ensemble trees to avoid under and overfitting
- Built with 100 trees, 15 max depth, 0.8 sample rate, and 2 folds for cross-validation
- One issue that I noticed was the speed/computational resources utilized for training this ensemble model

Gradient Boosting Machine (GBM)

- **Decision:** Another form of an “ensemble” model, but it focuses on remedying the mistakes of the previous tree instead of a simultaneously training the trees. Reduced the chance of overfitting
- Built with 500 trees, 0.01 learning rate, max depth of 7, and cross validation
- Utilized early stopping based on RMSE

XGBoost

- **Decision:** Typically performs the best due to its ability to avoid/reduce under- and overfitting. It was also cross-validated and trained using optimized hyperparameters.

Tools & Technologies

- H2O Machine Learning Library
- H2O Models – `h2o.randomForest`, `h2o.gbm`, `h2o.xgboost`

Stacked Ensemble Modeling

Requirements

- Combine the base learners into an ensemble model utilizing a neural network as the metalearner, which instructs H2o to use a deep learning model

Implementation & Decisions

- Cross validate predictions
- A stacked ensemble model by combining predictions

- Utilized to grasp the best of each model—similar to the functionality of a Random forest model, but the meta learner is a Neural Network

Tools & Technologies

- H2o Stacked Ensemble functionality

Model Evaluation

Requirements

- Assess the model performance via RMSE

Implementation & Decisions

- RMSE was used as the evaluation tool
- Validation results from CV guided the selection of the model
- Model performance was compared across the base learners and ensemble

Tools & Technologies

- H2o evaluation metrics
- RMSE (Root Mean Squared Error)

Changes to the ML

In order to improve the performance of the previous ML models, I decided to implement random search on all of the hyperparameters for the base learners. While I considered using grid search, I didn't have the computational resources to run each base learner with grid search, so I decided to do a random search. With this, each model's performance improved with a reduction of about 5,000 to 10,000 in the RMSE score.